

Implementation and Evaluation of VIPRS-x: Joint Effect-Size Modelling across East Asian and European Populations

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1 Introduction

The rapid growth of large-scale biobank data with comprehensive genotyping and phenotyping efforts is motivating the development of statistical methods for phenotype prediction with single-nucleotide polymorphism (SNP) array data. These rich biobank data can also power analyses of complex traits and diseases, revealing highly polygenic architectures. The polygenic scores have clinical relevance with some disease and health conditions, especially in applications related to disease risk stratification and personalized medicine. For example, in the large-scale prospective cohort study of Chinese populations, Dai et al. generated a PRS and evaluated its utility and effectiveness in predicting lung cancer risk in Chinese populations.

Standard PRS methods formulate the problem of polygenic risk estimation in terms of a multiple linear regression framework. Genotyping arrays combined with imputation can accurately capture the genotype of an individual at millions of genetic markers. Addressing such high dimensional data of hundreds or thousands of individuals faces computational and statistical challenges. Moreover, most individual-level GWAS data sources are protected for privacy concerns. Therefore, it's necessary to develop PRS methods that estimate PRSs on the basis of GWAS summary statistics. Bayesian PRS models not only can provide meaningful estimates of parameter uncertainties but also perform competitive predictive ability. However, a major limitation of existing Bayesian methods is that their scalability is affected by slow and inefficient inference techniques. One of the reasons for this limitation is that most Bayesian PRS methods implement MCMC algorithms to approximate posterior for the effect sizes. An alternative scheme for approximating the posterior density for the effect sizes is variational inference. The key point of Bayesian variational inference is the posterior distribution over a set of unobserved variables is approximated by a variational distribution which has a much simpler form than the posterior distribution.

Zabad and Li et al., (2023) [ZGL23] presented a variational inference of polygenic risk scores(VIPRS), a Bayesian summary statistics based PRS method that utilizes Variational Inference to approximate the posterior for the effect sizes. The paper showed that VIPRS is a scalable and flexible method that not only has the speed of heuristic approaches, but also performed good predictivity. For flexibility, it was tested by predicting with different familiar of priors on the effect size, paired with four distinct strategies for tuning the hyperparameters of the model. The scalability was demonstrated by evaluating the predictive accuracy of VIPRS with approximately 9.6 million SNPs. The project will implement VIPRS-x, which is modified from VIPRS and

aimed at tackling the problem of phenotypic prediction in structured populations or groups with heterogeneous ancestries. The essential component of this work is to jointly model effect sizes across different groups. The results will be compared with the benchmark work VIPRS applied on each subpopulation.

2 Materials and Methods

2.1 Simulated Dataset

The 1000 Genomes Project (1000GP) was designed to provide a comprehensive description of human genetic variation through the sequencing of multiple individuals. Over the three phases of the 1000GP, 2504 individuals across 26 populations were sequenced. These populations were classified into 5 major continental groups: Africa (AFR), America (AMR), Europe (EUR), East Asia (EAS), and South Asia (SAS). In the project, Europe and East Asia populations are selected. In each population, 100 SNPs from chromosome 22 are chosen. The phenotype is generated from $\mathbf{y}_p = \mathbf{X}_p \beta_p + \epsilon_p$ once the effect sizes are decided. When implementing the algorithm, I used the following relationships between $\mathbf{X}_p$ and $\mathbf{Y}_p$ to simplify the code programming:

$$\begin{aligned} \mathbf{X}_{j,p}^T \mathbf{X}_{j,p} &= N_p \\ \mathbf{X}_{j,p}^T \mathbf{X}_{m,p} &= N_p R_{jm,p} \\ \mathbf{y}_p^T \mathbf{X}_{j,p} &= N_p \hat{\beta}_{j,p} \end{aligned}$$

2.2 Methods Description

2.2.1 Baseline VIPRS

The dependence of phenotype on the genotype is modelled via the standard linear model

$$\mathbf{y} = \mathbf{X}\beta + \epsilon \quad (1)$$

where $\mathbf{y}$ is a $N \times 1$ vector of phenotypic measurements for N individuals, $\mathbf{X}$ is the $N \times M$ genotype matrix, β is the vector of effect sizes for each of the M markers and will be estimated. In the Bayesian framework, the problem is addressed by imposing a prior distribution over the effect sizes and then deriving a solution for the posterior distribution.

$$p(\beta|\mathbf{X}, \mathbf{y}, \theta) \propto p(\mathbf{y}|\mathbf{X}, \beta, \theta)p(\beta, \theta)$$

where θ includes all hyperparameters in the model. The spike-and-slab prior is assigned on the effect sizes $\beta_j \sim \pi N(\beta_j; 0, \sigma_\beta^2) + (1 - \pi)\delta_0$. In the model inference, the technique, of variational inference, is utilized. The essential component is to minimize the divergence between the true posterior densities and the approximated simple parametric distribution $q(\beta, \mathbf{s})$.

$$q(\beta, \mathbf{s}) = \prod_{j=1}^M q(\beta_j, s_j) = \prod_{j=1}^M N(\beta_j; \mu_j, \sigma_j^2) \text{Bern}(s_j; \gamma_j) \quad (2)$$

where μ_j , σ_j^2 , γ_j are the variational parameters. When minimizing KL-Divergence, or maximizing ELBO, the EM algorithm is used. In the E-step, the variational parameters will be updated for each SNP j . In M-step, the hyperparameters σ_β^2 , π , σ_ϵ^2 are updated then. ELBO will be implemented to check if it is converged for some tolerance.

2.2.2 VIPRS-x

The VIPRS-x is aimed to address the problem of phenotypic prediction in structured heterogeneous populations or groups. Therefore, the equation (1) will be modified to

$$\mathbf{y}_p = \mathbf{X}_p \beta_p + \epsilon_p; p \in \{1, \dots, P\} \quad (3)$$

In order to model the effect size across populations jointly with a multivariate Gaussian mixture density, equation (2), will be modified to

$$q(\beta, \mathbf{s}) = \prod_{j=1}^M q(\beta_j, s_j) = \prod_{j=1}^M q(\beta_j | s_j) q(s_j) = \prod_{j=1}^M \left(\prod_{k=1}^K N_p(\beta_j; \mu_{jk}, \Sigma_{jk})^{s_{jk}} \left(\prod_{k=1}^K \gamma_{jk}^{s_{jk}} \right) \right) \quad (4)$$

where μ_{jk} , Σ_{jk} , γ_{jk} for all j and k are the variational parameters that require tuning. μ_{jk} is a P vector and Σ_{jk} is a $P \times P$ matrix, where P is the number of populations. Therefore, similar to Section 2.2.1, in the E-step, μ_{jk} , Σ_{jk} , and γ_{jk} will be updated correspondingly. Hyperparameters Σ_k , $\sigma_{\epsilon,p}^2$, and π_k will be updated in the M-step.

Note: In the project, $K = 2$ and $P = 2$. As expected, when we implement the E-step, μ_{jk} and the diagonal elements in Σ_{jk} are supposed to be close to zero. u_{jk} in equation (11) from the project description should be approaching $\log(\pi_k)$.

3 Results

3.1 Pearson Correlation

3.1.1 VIPRS-x

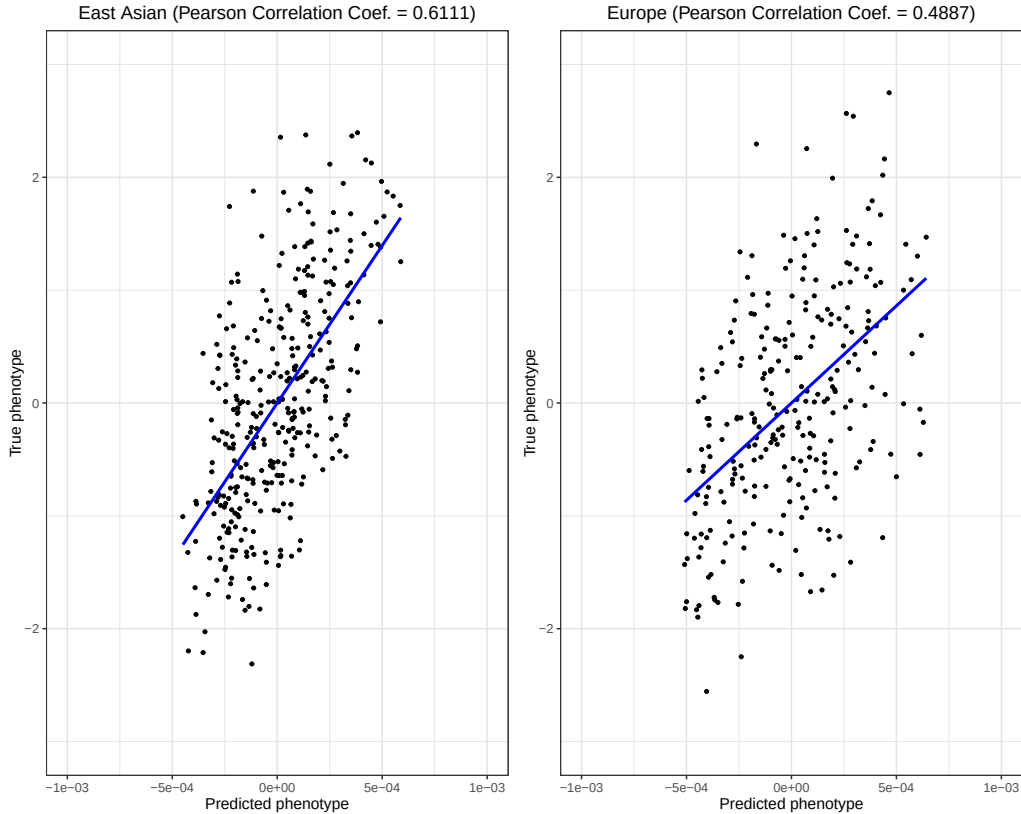


Figure 1: Person Correlation on Train Dataset

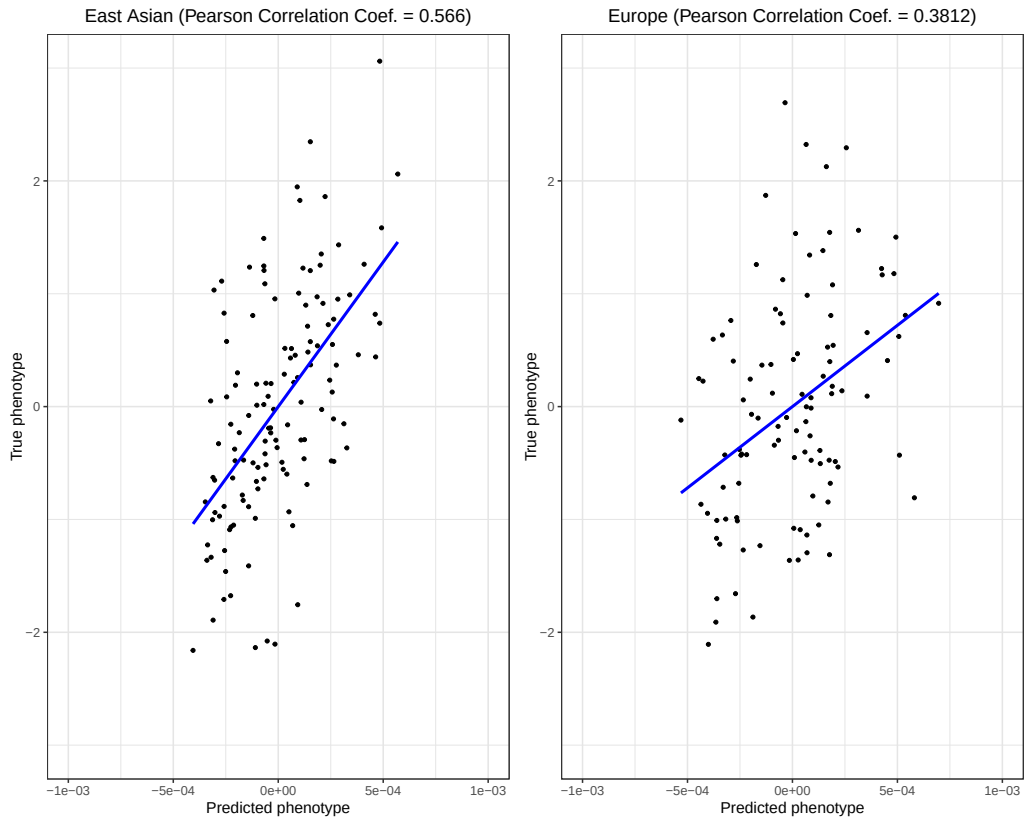


Figure 2: Person Correlation on test Dataset

3.1.2 VIPRS

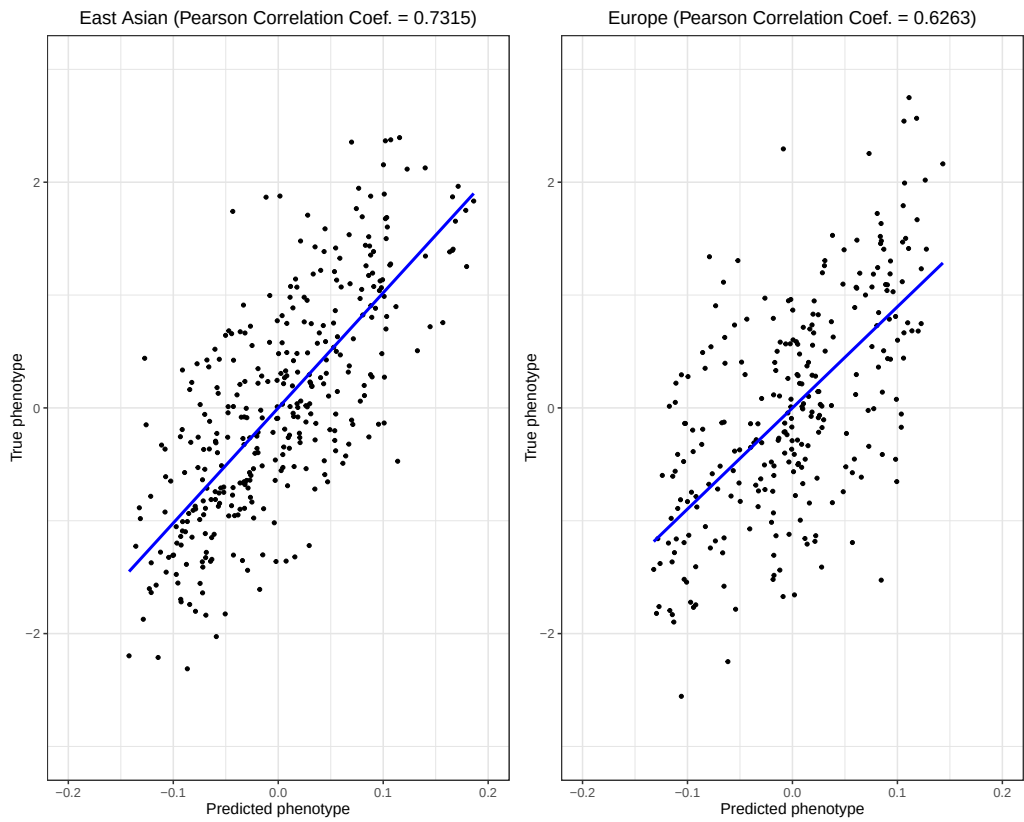


Figure 3: Person Correlation on train Dataset

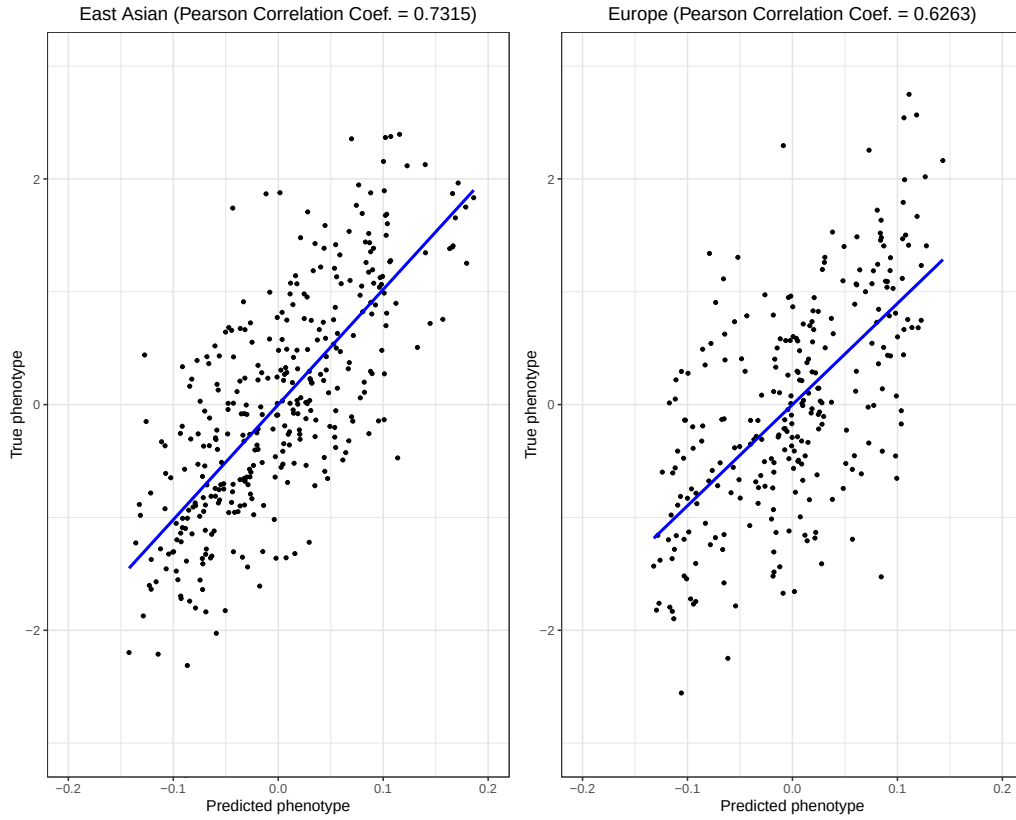


Figure 4: Person Correlation on test Dataset

3.2 Squared Errors

The squared error is calculated as follows:

$$\text{error} = ||\hat{\mathbf{y}}_{\mathbf{p}} - \mathbf{y}_{\mathbf{p}}||_2$$

VIPRS		
Populations	Train	Test
East Asian	17.7760	10.7644
Europe	15.7962	9.8137

VIPRS-x		
Populations	Train	Test
East Asian	18.7325	11.3123
Europe	16.4901	10.19701

4 Discussion

As we see from the Person correlation in Section 3, VIPRS performs better than VIPRS-x in both training data and testing data. However, in terms of squared errors, these two methods do

not show significant differences. One of the reasons is probably the setup of covariance-variance for two populations. In the joint modeling framework, the effect sizes $\beta_{j,p}$ are estimated simultaneously, which is corresponded to $N_p(\beta_j; \mu_{jk}, \Sigma_{jk})$ in equation (4). In this case, the covariance between two populations also needs to be estimated.

In VIPRS-x, the Pearson correlations in the European population are both lower in training data and testing data. It means the predicted phenotypes are pretty underestimated. The possible reason is $\mu_{jk,p}$ is not estimated properly in the implementation because the $\sigma_{\epsilon,p}^2$ is not updated well.

Implementation details: the initialization follows the suggestions in the Project Description. To enjoy the benefits of the standardized dataset, $X_{j,p}^T X_{j,p}$, $X_{j,p}^T y_p$, and $X_{j,p}^T X_{m,p}$ will be replaced by N_p , $N_p \hat{\beta}_{j,p}$ and $N_p R_{jm,p}$ respectively. The ELBO is also implemented to check the convergence. $\sigma_{\epsilon,p}^2$ does not need to be updated at first, which can be fixed at a reasonable value, such as 0.90. After checking the convergence of ELBO, $\sigma_{\epsilon,p}^2$ can be updated. When updating $\sigma_{\epsilon,p}^2$, $\sum_{j=1}^M X_{j,p}^T X_{j,p} \text{Var}_q(\beta_j, p)$ will be replaced by ζ_j , where $\zeta_j = X_{j,p}^T X_{j,p} \text{Var}_q(\beta_{j,p})$. In VIPRS-x, after 30-40 iterations, ELBO will be converged if the tolerance is 10^{-4} . In VIPRS, for East Asian population, ELBO will be converged after 7 iterations; for Europe population, the iterations will be 64.

In the M-step, hyperparameters $\pi_k, \Sigma_k, \sigma_{\epsilon,p}^2$ will be updated to maximize the surrogate objective. It works well in many settings, but it is also easy to get the local optima, not global one. Therefore, when tuning hyperparameters, some other alternative strategies can be explored: Grid Search, Bayesian optimization, and Bayesian model-averaging. But it might be difficult to implement grid search in joint modelling framework since the hyperparameter Σ_k is a matrix. Bayesian model-averaging is more likely to be practiced comparing with two other strategies. Instead of fixing π_k to particular values, the VIPRS-x will be fitted along a grid of π_k values and the other two hyperparameters $\Sigma_k, \sigma_{\epsilon,p}^2$ will be updated in use of the closed form in M-step.